

Laerke.eu BETA boards guide

Version: 2.04 Blink & bin-file local install.

Updated: mmbo 2026-07-07

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Welcome

Hi

Here is the assembly and user guide for the Laerke.eu pressure sensors.

They can be used in two different ways. 1) by using a microcontroller board or 2) connected to Pioreactors.

If you have the laerke.eu board, then you should soon receive your user information in an email. If you want to use the sensors with Pioreactors, then first assemble them, and then see the section on connecting them to Pioreactors.

Using the microcontroller and the laerke.eu/data server. You should receive the following info:

Database user email: "User-e-mail"

password: "passpasspass"

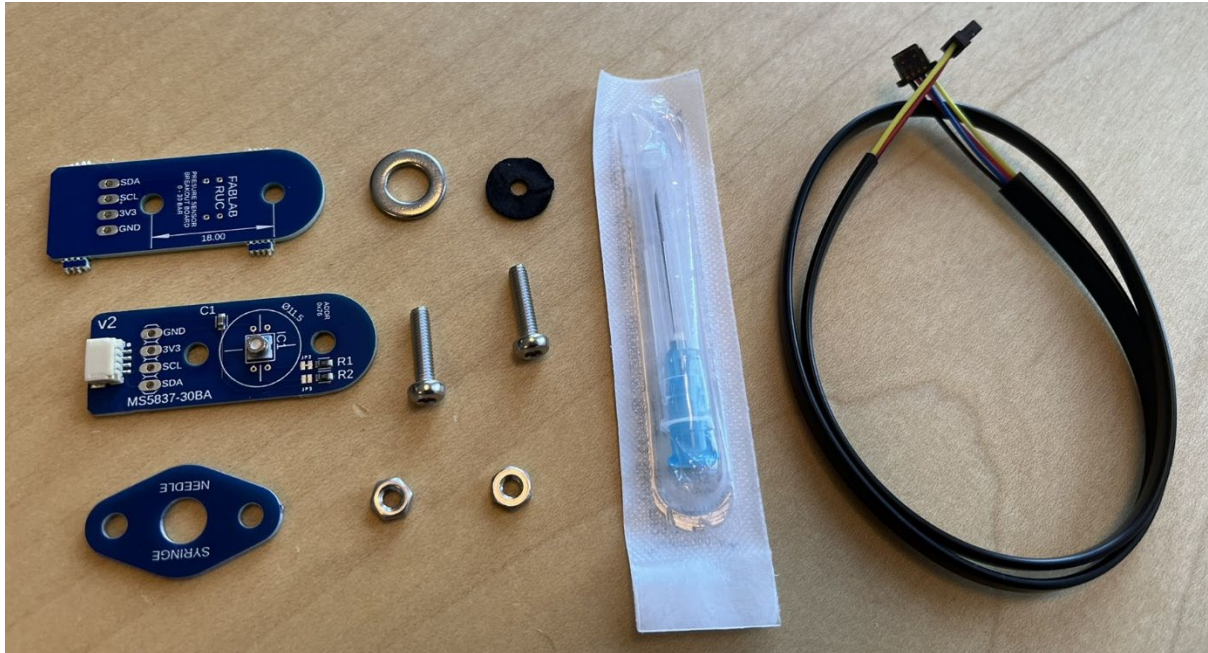
User UID: "TG5Wg4rtfHQzn5XOBj7nczYbxsO2"

What you now need to do is:

1. Assemble the sensors
2. Write Wi-Fi & user info onto the microSD card
3. Load the firmware/software code onto the microcontroller

4. Connect the board to power; it is now logging data.

Assemble the single sensors



Parts needed

Provided:

- Received Email with database user email, PW, and UID
- Kit with parts.

You should provide

- microSD reader/writer
- USB c cable
- Wi-Fi that allows the microcontroller to log on.
- Cable binders/cable ties.
- Small stickers or tape



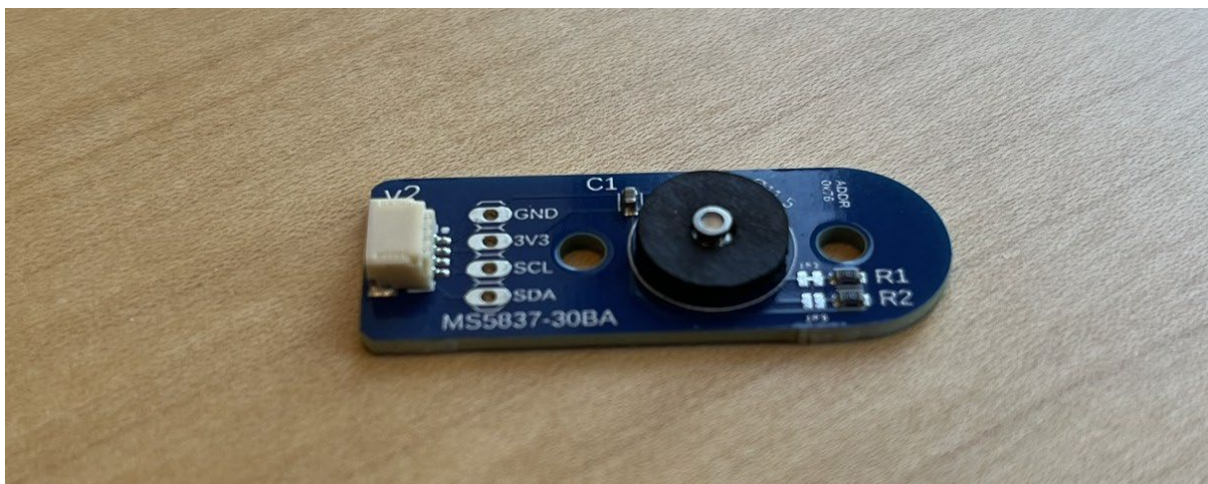
1. Connect the needle

- Add the washer on the needle.
The rounded sides of the washer should face the needle flange, and the flat side should be against the needle holder PCB.
- Add the needle holder
- Reconnect the needle safety cover
- Add bolts/fasteners



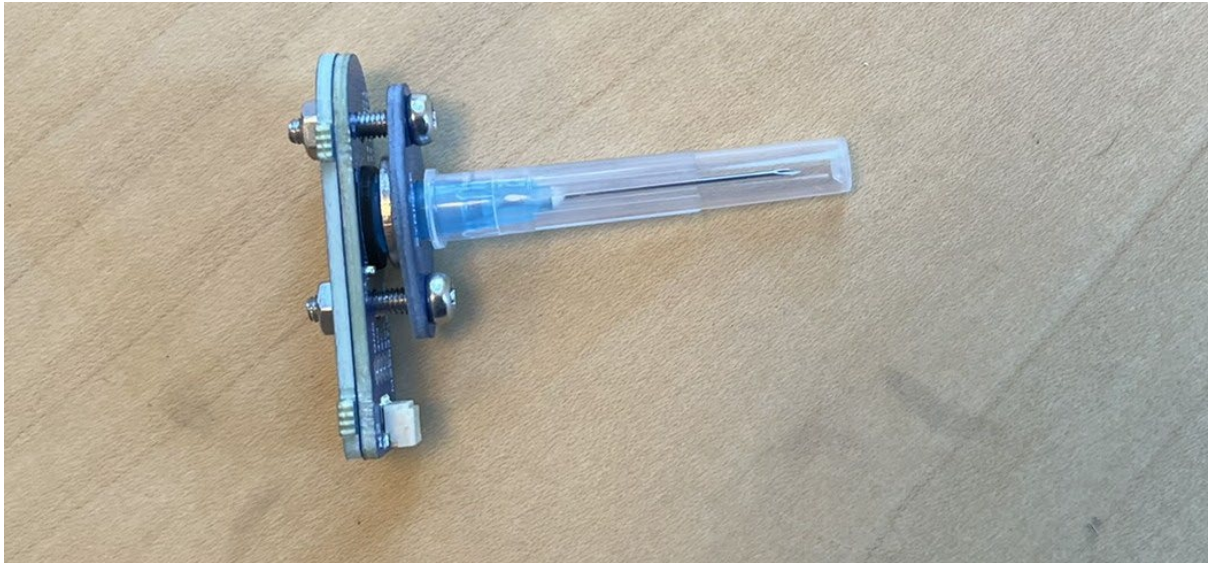
1. Wash the gadget / O-ring

- Use a wet and a dry tissue
- Rub it between the fingers and inspect. Carefully remove all sod, dust, paper



2. Add the gadget to the sensor PCB

- Do not damage the actual sensor, it's fragile and can break off. Also, do not touch the top white part of the actual sensor.



3. Assemble

- Connect the needle and top plate to the sensor PCB
- Add the back-plate. (pcb without electronics)
- Add the bolts
- Jitter around slightly such that everything aligns properly.

Things don't align?

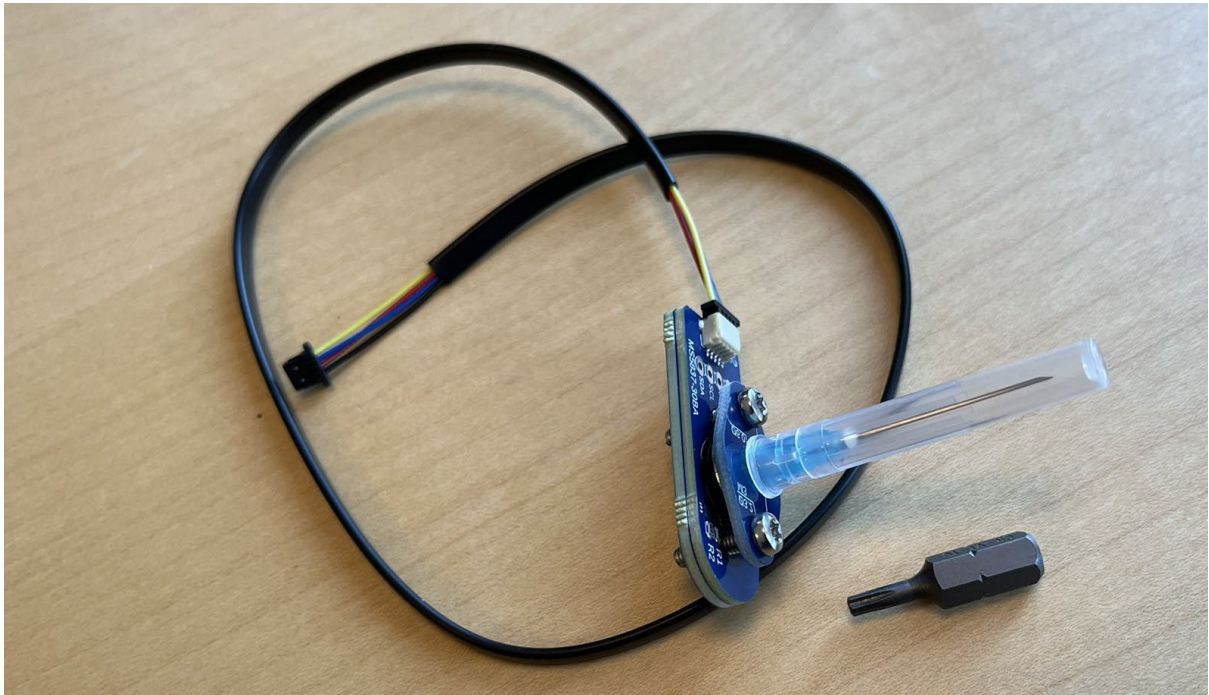
- Depending on the needle type and size, this size of washer might not be optimal. If this setup doesn't work with your needles. You can try removing the washer or find other needle sizes.



4. Tighten the bolts

- Use your fingers
- Shift between the bolts to make it nice and even.

- You can tighten it with your fingers as much as you can. This should make it tight without breaking it. You don't need to use larger tools to make it tight.

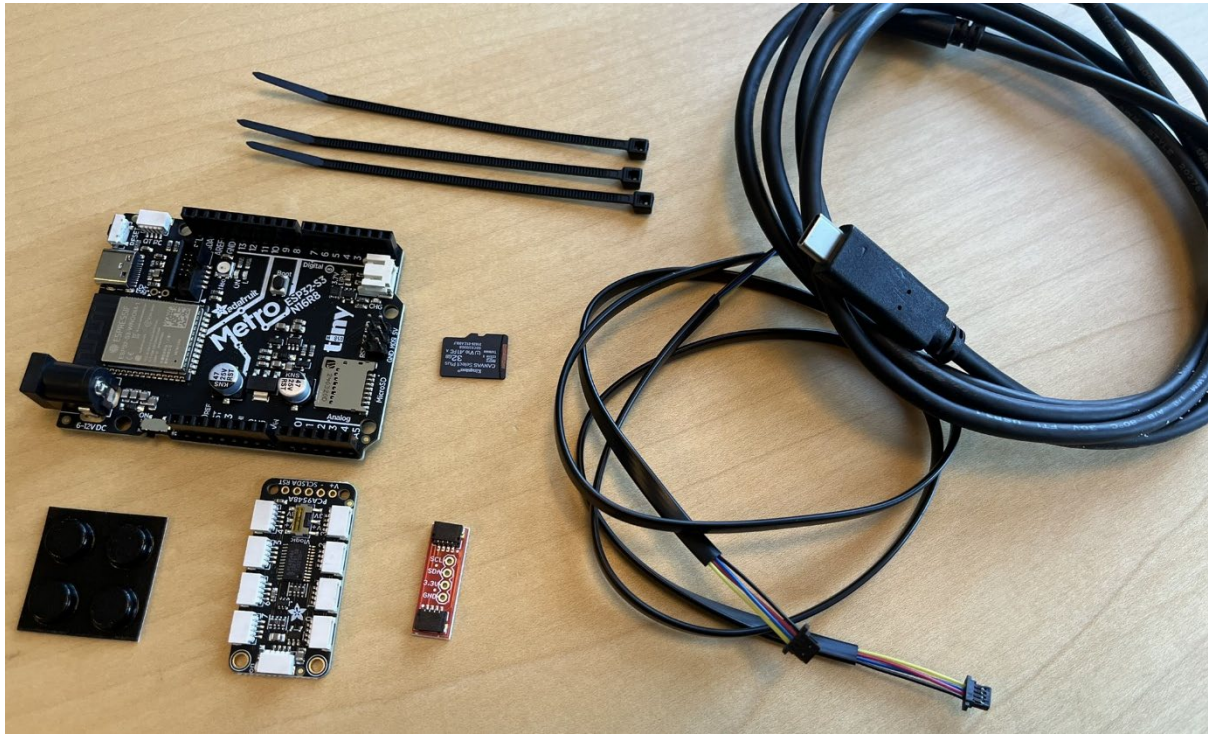


5. Connect the cable.

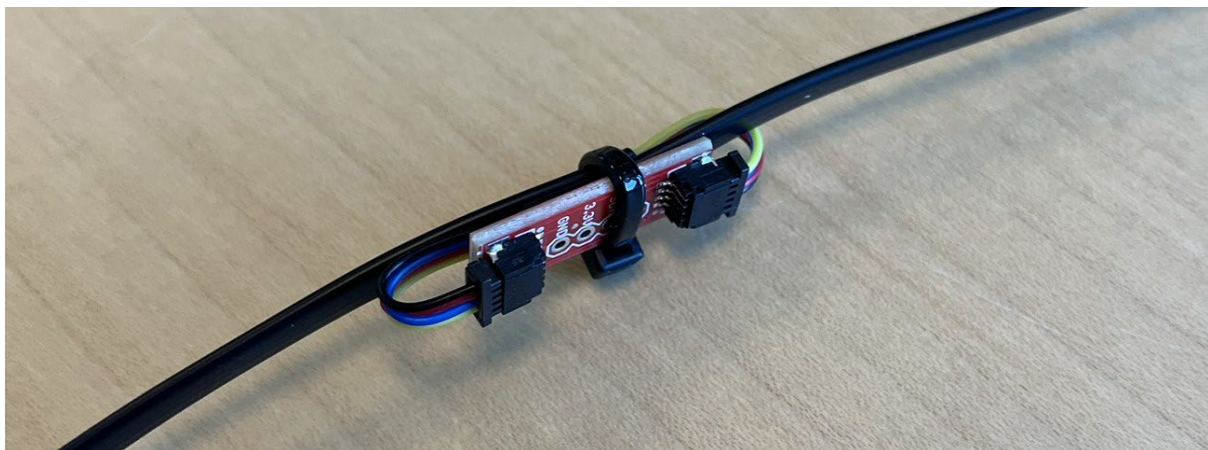
- And you're done.

Laerke Sensor Board Guides:

1. Assemble the beta 8 sensor system

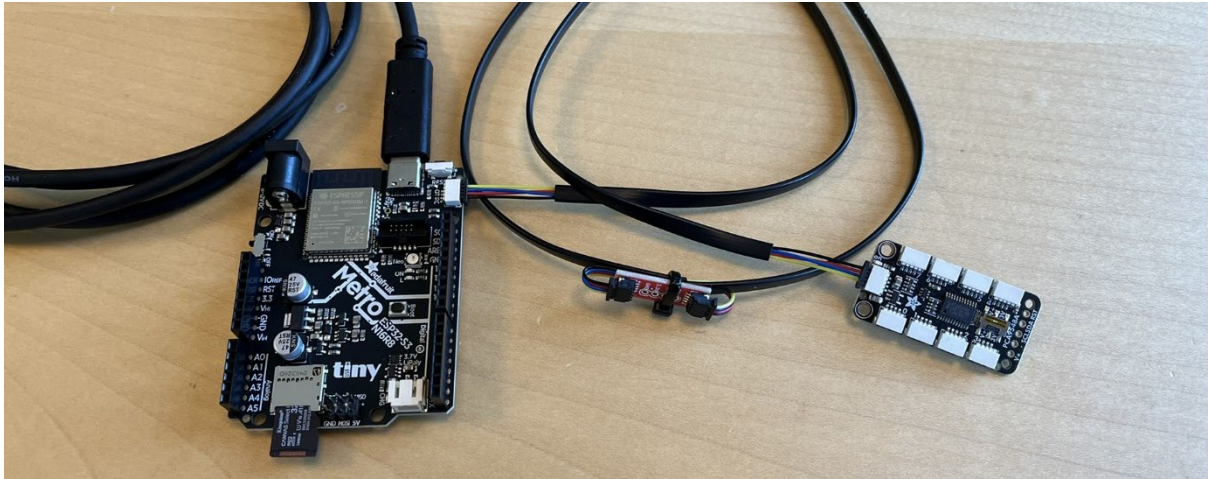


Parts needed (Photo above)



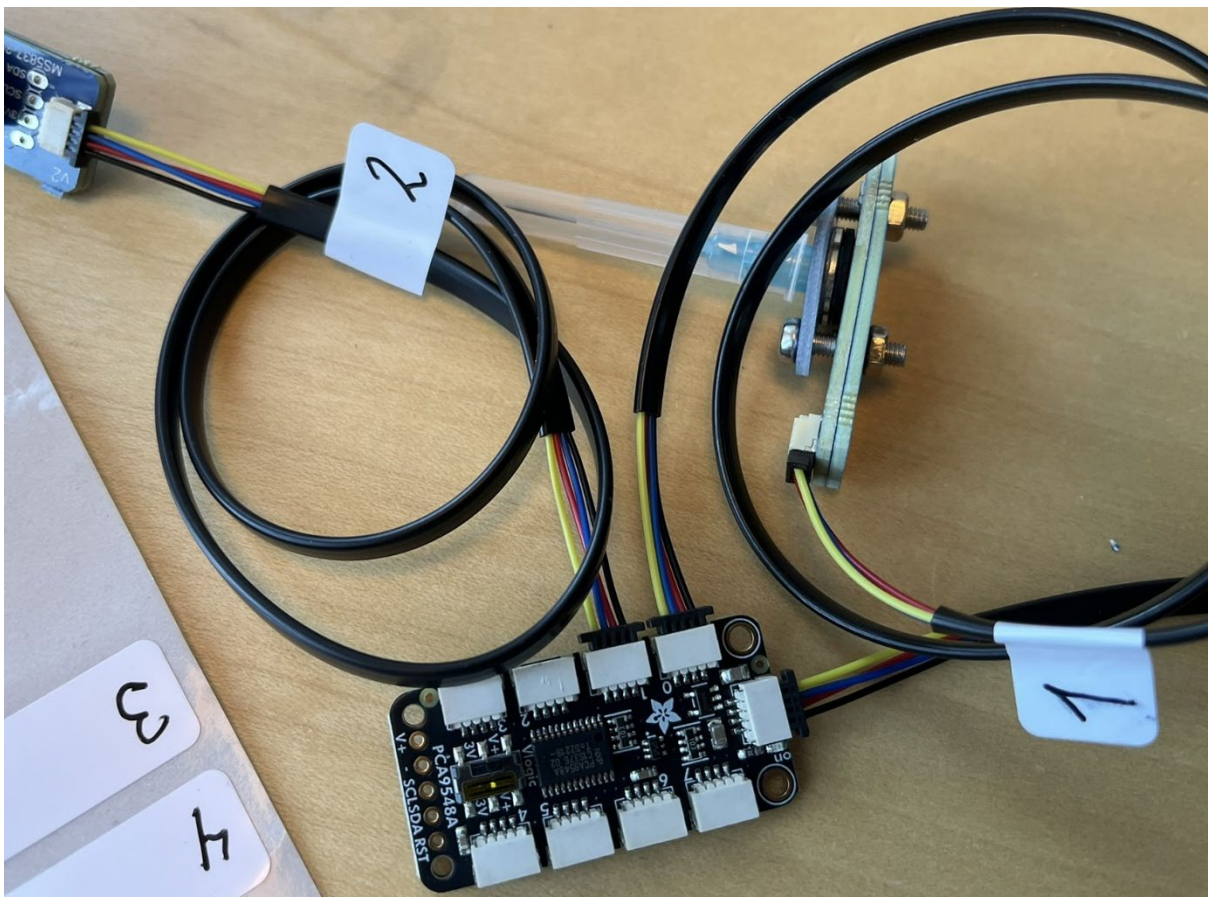
1. Connect two wires

- Connect two cables/wires to the red dual connector
- Turn the wires 180° and fix them with a cable binder such that the cables don't move



2. Connect the multiplexer and microSD

- Connect the multiplexer with the microcontroller
- Add the microSD card
- Add USB-C cable



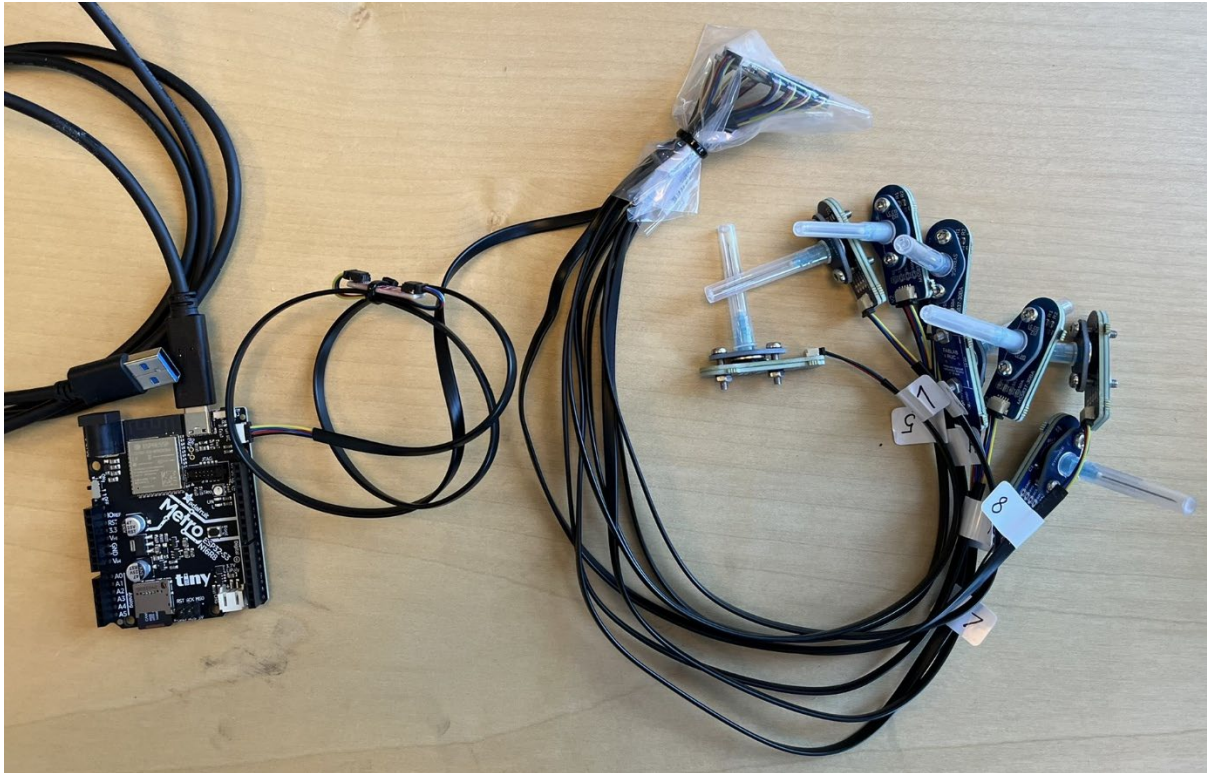
3. Label and add sensors

- Add sensors to the multiplexer
- Label the cable near the sensor.
 - Multiplexer 0 = sensor 1
 - Multiplexer 1 = sensor 2, ect. Ect.



4. Secure wires

- Cut a plastic bag and add it on top of the multiplexer
- Fix and secure the wires with a cable binder.
- All sensors do not need to be connected at the same time for things to work. If the number of sensors used changes. You can always disconnect or connect as required to remove the clutter of unused sensors.



5. Done

2. Write user data to SD-card

1. Safe user data on to SD card

The user, Wi-Fi and sensor ID information is written onto the microSD cards for each sensor.

It is stored in a text file called “config.json”. See an example below.

- The measurement interval units are seconds. A good standard is a reading every 3 minutes, approximately. 120 seconds.

Steps.

1. Create a standard .txt file and copy the text below. Update it to fit your Wi-Fi and user data.
2. Save it to the microSD card, naming it “config.json”

```
{"microcontroller-name":"Sensor_01",
```

```
"measurement-interval":120,
```

```
"save-to-sd-card-only":false,
```

```
"wifi-ssid":"Mobile_hotspot_name",  
"wifi-password":"Wi-Fi_password_pass_pass",  
"user-email":"mmbo@dtu.dk",  
"user-password":"Data_base_password",  
"user-id":"TG5Wg4sgeHQzn5XOBj7nczYbxsO2"}
```

Notes:

- Microcontroller name: If you have more systems. Change each name to fit the use, the lab, etc.
- Measurement intervals are seconds between readings.
- Wi-Fi SSID is the Wi-Fi name

3. Wifi

Setup a local wifi

The sensors need to log on to a local Wi-Fi before they can upload data to the online server.

1. Test the system by sharing the local Wi-Fi hotspot on your phone. And write that data to the config.json wifi info.
2. Change to a permanent wifi.
 - a. We recommend a stand-alone mobile hotspot with a data simcard. (see below) that will work.
 - b. MAC-address. If you are using an institutional wifi. The IT department might want to get the MAC address of each board and add it to their system.

Stand-alone mobile data router

We recommend a standalone mobile data router as the local wifi hotspot.

Something like this: <https://www.tp-link.com/nordic/home-networking/mifi/m7350/>

In theory, they can connect to any Wi-Fi at universities or other institutions. This might require extensive configuration and interaction with the IT department due to security settings.

Sometimes the unique microprocessor number (MAC) needs to be typed into the system.

Getting the board MAC address with Android phone

You can see the board's MAC address on the device it logs into.

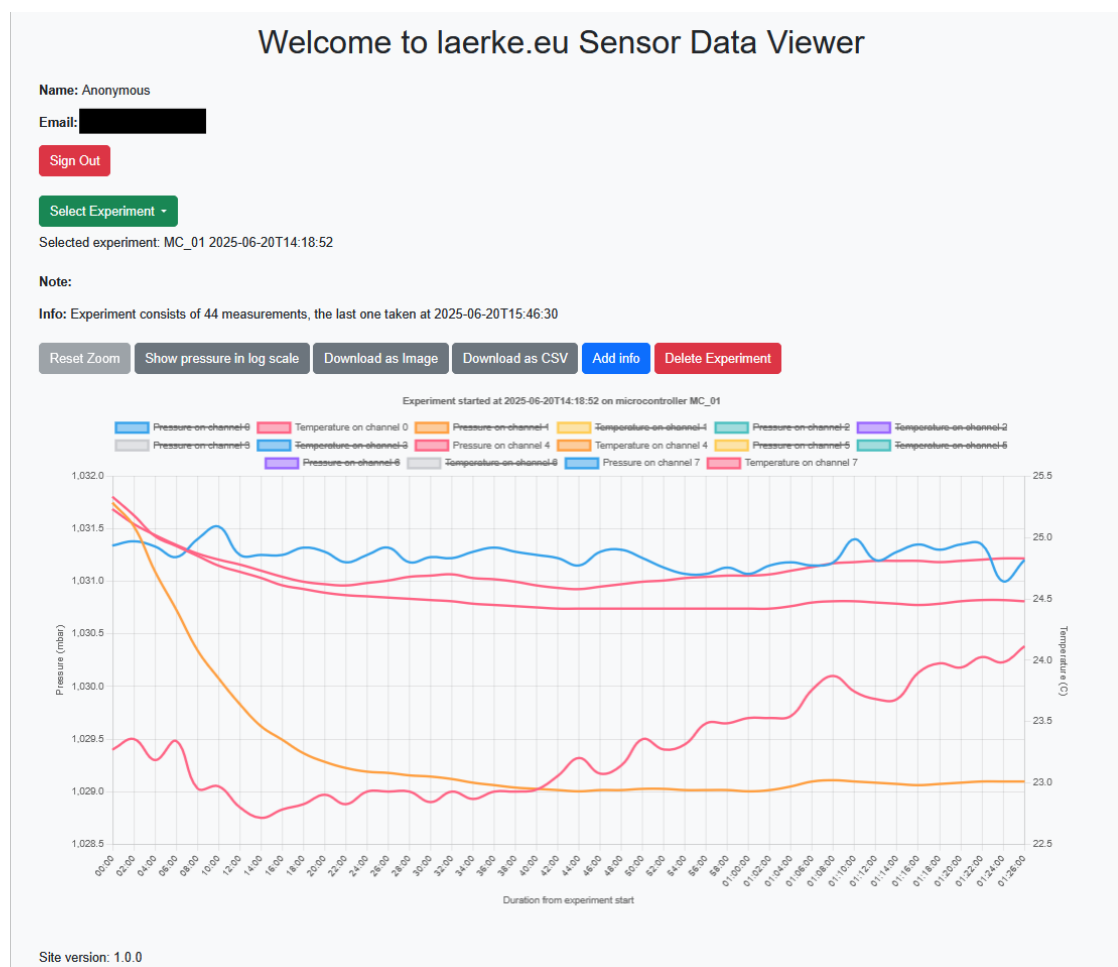
1. On an android device, create a wifi hotspot.
2. Write the wifi name and password to the config.json file and save on the SD card.
3. When the board have logged on to the wifi. You should be able to see what device that have locked on and their MAC address
4. Give this MAC address to the IT department, such that they can put the board on the "safe users" list.

Unfortunately, this does not work on iPhone.

4. Start Experiment, view and download data

1. Connect to power to start / restart logging

- Connect the sensor to power to start the experiment
- When the power is connected a datafile is created on the SDcard with the amount of sensors, and each data point is logged with a time-stamp. The data is then uploaded to the data server for online viewing.
- Disconnect and reconnect to start a new experiment and logging file
- If you change the amount of sensors, restart the microcontroller. It will then re-detect how many sensors that are connected and start the logging of a new experiment.



2. View data on-line:

1. Go to laerke.eu / data
2. Type in user name: "user@university.eu"
3. Type in firebase password: "your-unique-passpasspass"
4. Select the experiment sorted by sensor name + date-time to view data

3. Download data

- Follow the log-in above
- Select the right experiment
- Click “Download as CSV”

5. Blink-messages & errors

- Initiating and connecting to all systems a couple of times.
- If failing to connect to WiFi, it will save locally to the SD-card

Yellow – Power on

- x2: Power on or reset initialization

Green - Initialize

- 4x Green: All systems initialized and ready for main sampling loop
- 5x Green: SD Card Initialized
- 3x Green: WiFi Connected
- 2x Green: Write SD-card success

Purple: Running

- 2x Purple: DB Write Success
- 1x Purple: Sensor Added

Red - Errors

- 2x RED: WiFi Fail
- 3x RED: DB Write Fail
- 5x RED: SD Card not detected
- 6x RED: SD Write Fail
- 8x RED: No Sensors Detected

6. Update the board firmware code

The code running on the board can be updated. As we develop the project new versions of the code will be released. It can be “push” updated “over the air” by us or locally by you connecting to a computer.

Update Metro ESP32-S3 board using Adafruit WebSerial ESPTool:

- Obtain the relevant “Pressure.ino.bin” file with the hardware code.
- 1. Connect the board to a computer
- 2. Enter ROM bootloader mode on the board:
 - 1) Press and Hold Boot button. 2) Press and Release reset button. 3) Release boot button (The cycling colour now stays constant). If needed, see Step 2 here: <https://learn.adafruit.com/adafruit-metro-esp32-s3/factory-reset#step-2-enter-rom-bootloader-mode-3106832>
- 3. Load code to board:
 - 1. In Chrome go to https://adafruit.github.io/Adafruit_WebSerial_ESPTool/
 - 2. Press connect on the Adafruit WebSerial ESPTool page and select the board. It will be called something like USB JTAG/Serial debug unit (COM8) paired
 - 3. In the first file selector. Select the provided Pressure.ino.bin file and set Offset to 0x10000
 - 4. Press Program and wait for it to finish.
 - 5. Reset:
Press the Reset button on the board or disconnect it from the computer and reconnect it to power.
- 4. The Board is now updated.
 - Update board and wifi information on the SD-card if needed.
 - Run a test.
 - Start experiments.

7. Troubleshooting

Standard procedure

1. Restart the board. Remove power and reconnect
2. Check **ALL** cables:
 - a. Are any of the pins inside the connectors bent or broken?
 - b. Did the entire connector loosen from the PCB?
 - c. Are all the cables properly attached? They can easily come off or be loose when handling, especially if you have shaking incubators.
3. Inspect the board and sensor PCBs visually
 - a. Any salt, dirt, water that disrupts or short-circuits things? -> disconnect power and clean things up -> Reconnect.
4. Change sensor PCBs connectors. Shift the connections on the multiplexer. Is it the sensor PCB or the wire connections?
5. Sensor not working
 - a. Remove the needle carefully not to break the actual sensor of the PCB
 - b. Visually inspect the sensor. Any dirt, did it disconnect? You should not touch or wash the white polymer protecting the sensor. The sensor will break if damaged.
 - c. Change the needle. It might be dirty
 - d. Let the PCB dry completely if humid
 - e. Reconnect everything and try again.
 - f. Sensor still not working... hmmm, it is probably broken. Order a new sensor. You can return it to us, and we would like to figure out what went wrong.

Connect to Pioreactors

The screenshot shows the Pioreactor web interface. On the left is a sidebar with navigation links: Overview, Pioreactors, Profiles, Configuration, Inventory, Calibrations, Export data, Past experiments, and Plugins (which is highlighted). The main content area displays a list of plugins. The 'laerke.eu-pressure-reading-plugin' is highlighted with a yellow box. It is described as 'A pressure sensor system for online tracking and analysis of gas formation or consumption in vials.' and shows a status of 'Installed on Leader01'.

Plugin Name	Description	Status	Action
Pioreactor	Calibrate your LEDs using an external light probe. Also comes with a Day/Night LED automation to use your new calibrated LEDs.	Install on Leader01	View
co2-reading-plugin	Retrieve CO2, temperature, and humidity readings from an attached Adafruit SCD30 or SCD40/SCD41 sensor.	Install on Leader01	View
spectrometer-reading-plugin	Retrieve 415nm, 445nm, 480nm, 515nm, 555nm, 590nm, 630nm, and 680nm readings from an attached Adafruit AS7341 sensor.	Install on Leader01	View
pioled-display-plugin	Display growth rate, normalized OD, and temperature for each Pioreactor using the Adafruit PIOLED display.	Install on Leader01	View
pioreactor-basic-auth-for-ui	Adding basic auth to the Pioreactor UI.	Install on Leader01	View
pioreactor-calibration-shrinkage	Shrink multiple calibrations across workers in a Pioreactor cluster using multi-level modelling: better inference for free!	Install on Leader01	View
temperature-expansion-kit-plugin	Expand the Pioreactor's temperature performance. Additional hardware kit required!	Install on Leader01	View
laerke.eu-pressure-reading-plugin	A pressure sensor system for online tracking and analysis of gas formation or consumption in vials.	Installed on Leader01	View

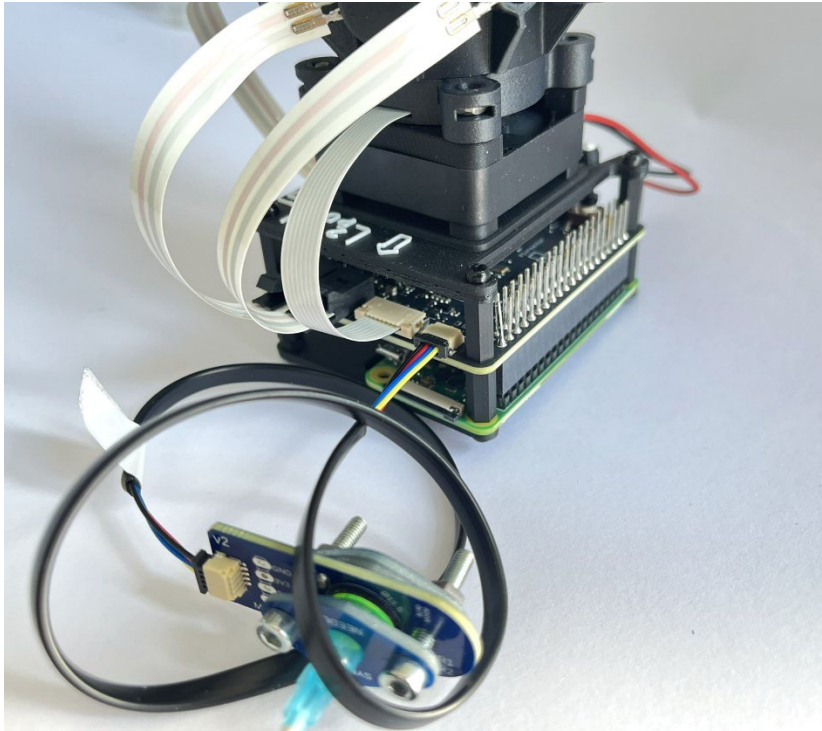
1. Install Laerke.eu Pioreactor plug-in

- In the Pioreactor interface go to plugins and install the “laerke.eu-pressure-reading-plugin”



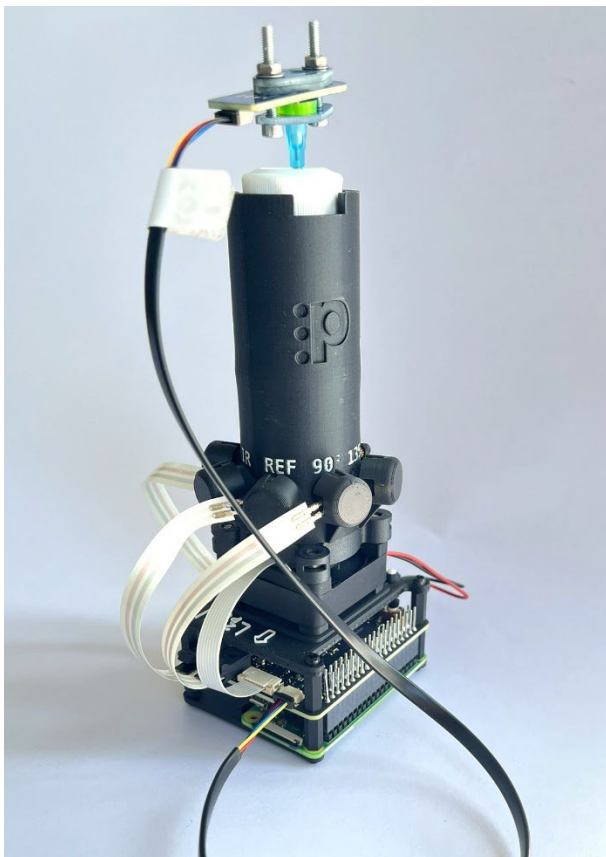
2. Add rubber stopper to caps

- Add the butyl rubber stopper to the white cap or a 3D-printed pioreactor cap
- Use a blond screwdriver or similar to push it properly down along the edges.

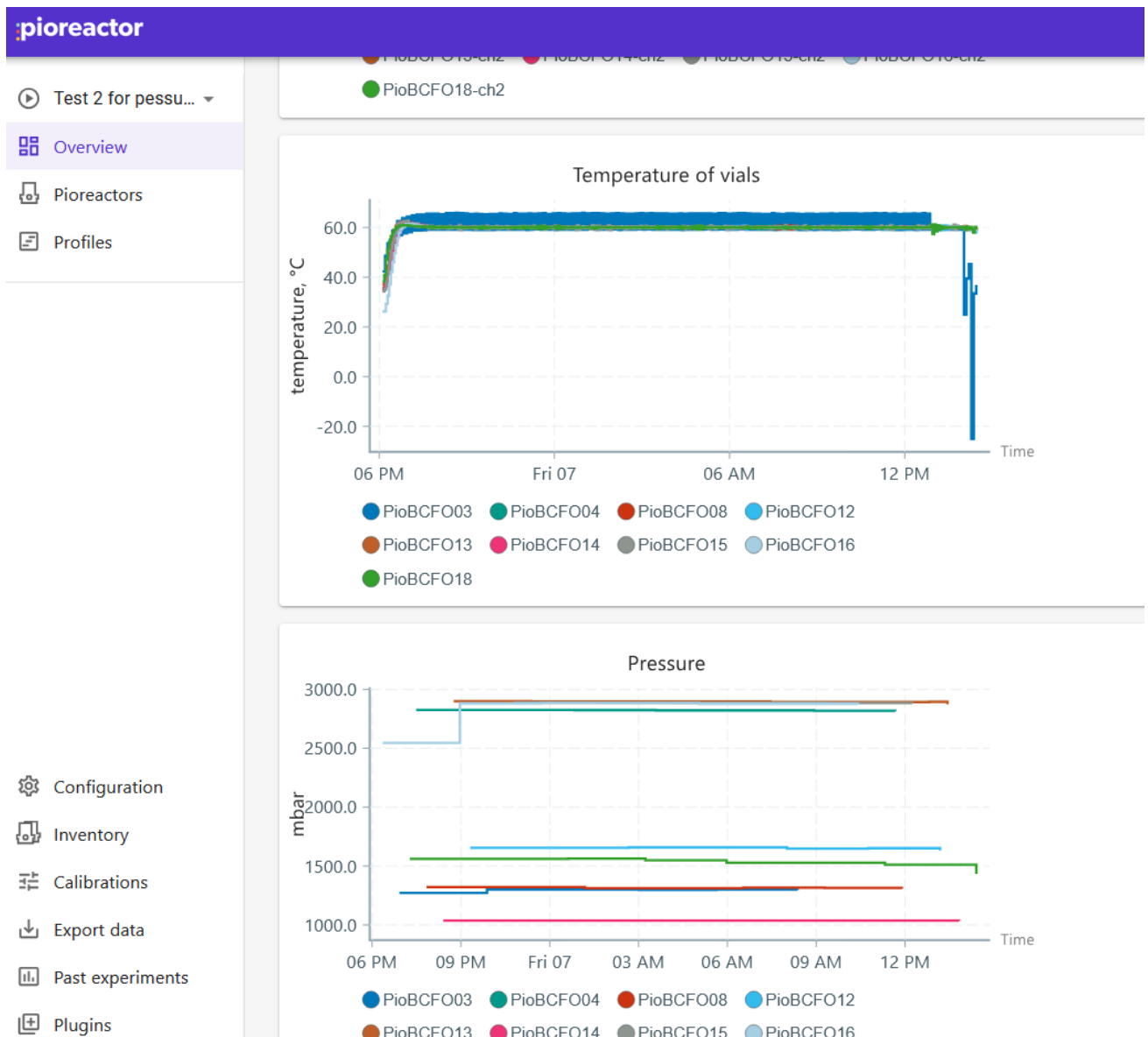


3. Connect to pioreactor HAT

- Connect the laerke pressure sensor to the pioreactor hat as shown



4. Push needle through rubber stopper



5. See the readings in the interface

- Look at the data for each sensor and experiment